Player-Managed Potion Shop

A shop-management game: ingredients become inventory

How much gold will you commit before opening?

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The opening batch

Buy ingredients before you know how busy the shop will be.

One ingredient-and-bottle kit makes one potion.
NPC customers buy from your opening batch.
A wholesaler buys leftovers at a discount at closing.

One opening batch. Closing clearance. No stock carryover.

Inputs

Ingredients + bottle: 30 gold. Sale: 50. Clearance: 10.

Shop days: Busy 35%, Normal 45%, Quiet 20%.
Demand outcomes: 40 to 100 potions.
Probabilities adapt the supplied course example.

Inputs adapted from the supplied course notes, pp. 68 and 75.

What the batch earns

Sold = min(stock, demand)

Cash = sales - production cost + closing clearance
Textbook score = cash - foregone margin on unmet demand
At stock 70 and demand 90: cash 1,400; score 1,000 gold.

The unmet-demand deduction is a scoring convention, not a cash expense.

Experiment and verification

30 runs of 1,000 shop days. Eleven batch quantities.

30,000 simulated shop days. 330,000 stock evaluations.
The same customer demand for each batch quantity.
Hand cases, source-example regression and exact expectation.

31 tests passed. Exact expectation checks the sampling result.

Stock comparison



Potions in the opening batch. Values average all 30 runs.

Conclusion

Stock 70 maximizes the textbook score.

Simulated score: 928.59 gold. Exact score: 931.00 gold.
Cash profit alone selects 60 potions.
Changing price to 80 gold changes the score winner to 80.

The price sensitivity test assumes unchanged customer demand.